

Obed Junias

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PROFESSIONAL SUMMARY

Generative AI Engineer specializing in **LLM systems, agentic workflows, and evaluation**. Experience building scalable ML/NLP pipelines and cloud-native services, with a strong foundation in production ML, and GPU-aware optimization. Interested in roles spanning **LLM serving, Agentic and RAG Systems, SFT/PEFT**, and **reliable LLM evaluation** for real-world deployment.

INDUSTRY EXPERIENCE

Senior Member of Technical Staff at Oracle Corporation

Aug 2021 - Aug 2024

- Designed and implemented scalable automation frameworks in Python, TestNG, and BATS, **reducing manual validation effort by 95%** across [Oracle REST Data Services](#) and [SQLcl](#).
- Built a Python-based migration framework to automate [Oracle APEX](#) application transfers, improving deployment reliability and supporting **200+ internal users**.
- **Led validation and release readiness** for a new [Oracle Cloud DB Tools](#) feature, coordinating a **15-engineer** effort and delivering production readiness in 2 months.

Machine Learning Engineer Intern at Hewlett Packard Enterprise

Feb 2021 - Jul 2021

- Built intelligent automation pipelines using Groovy and [WorkFusion](#) OCR to process unstructured documents, **improving processing speed by 40% and reducing errors by 80%**.

RELEVANT PROJECTS

- **Adapter-Based Multi-Agent LLM Serving System**
 - Built a **custom multi-agent LLM serving system** executing 14 task-specialized agents as dynamic **LoRA adapters** over a shared Llama-3B backbone, and **benchmarked against vLLM-style serving**.
 - Implemented **dynamic adapter loading, eviction, and scheduling** under strict **5–15 GB GPU memory budgets**, streaming adapters from CPU to GPU at runtime.
 - Evaluated high-entropy agentic workloads (up to **140 adapter switches per run**) across multiple schedulers, benchmarking **latency, throughput, GPU utilization, and switching overhead**.
 - Demonstrated that **LoRA adapters load in less than 1s** versus **9-22s full-model loads**, yielding **20-90x lower cumulative switching overhead** and enabling execution where full models are infeasible.
 - **Tech:** vLLM, PyTorch, HuggingFace Transformers, PEFT/LoRA, CUDA, Python
- **ScoutR: An Agentic Football Transfer Intelligence Platform**
 - Designed and shipped **ScoutR**, an **agentic AI transfer intelligence platform** that monitors player data across leagues and surfaces candidates aligned to club tactical and financial constraints.
 - Built a **multi-agent pipeline** where Gemini parses scouting requests into typed **Pydantic criteria**, then produces tactical-fit analysis, valuation summaries, and scouting outputs with schema enforcement.
 - Implemented **RAG with ChromaDB** over StatsBomb event data to ground recommendations in statistical evidence, reducing hallucinated comparisons and improving trust in transfer decisions.
 - Designed a **3-tier LLM fallback architecture** for graceful degradation during outages and rate limits, maintaining continuity for clubs without large analytics teams.
 - Engineered a **real-time streaming UX** using Server-Sent Events (SSE), exposing step-by-step agent reasoning so users can follow autonomous **long-horizon** decision workflows in one interface.
 - **Tech:** Gemini, Claude, LangGraph, Pydantic, ChromaDB, RAG, Prompt Engineering
- **GradCompass: Multi-Agent LLM Application Assistant**
 - Built a **production-grade multi-agent LLM system** for personalized graduate-application guidance, document review, and mock interview simulation.
 - Designed an **8-agent architecture** with role-specialized agents and a centralized orchestration layer.
 - Implemented **agentic execution and communication patterns** (tool use, routing, shared memory) and integrated a **Retrieval-Augmented Generation (RAG)** pipeline to ground recommendations.
 - Implemented agent coordination and response synthesis for coherent outputs across multi-step workflows.
 - **Tech:** Python, LangChain, LLM APIs, RAG, Vector Databases, Multi-Agent Orchestration

• **End-to-End Text-to-Comic Generation Platform**

- Built a **production-grade, cloud-native AI system** that converts natural-language narratives into comic strips using **LLMs and diffusion models**, with a **React frontend** and **Node.js backend**.
- Designed a **modular, asynchronous pipeline** deployed on **GKE**, using **Pub/Sub** for orchestration, **Cloud Storage** for persistence, and **Redis** for hot-path caching.
- Implemented **OAuth 2.1-based authentication** and per-request isolation, supporting **100+ users** and sustaining **20–30 concurrent generation jobs**.
- Reduced median end-to-end latency by **35–45%** through caching, pipeline parallelism, and reuse of intermediate artifacts under concurrent load.
- **Tech:** React, Node.js, Python, GKE, Pub/Sub, Redis, Cloud Storage, GPT, Stable Diffusion

• **Lost in Plot: Contrastive Learning-Based Movie Retrieval**

- Built a semantic retrieval system for **“tip-of-the-tongue” movie search**, handling vague natural-language queries, and constructed a **100K+ example synthetic dataset** for training and evaluation.
- Fine-tuned a **BERT-based dual encoder** using a **multi-task contrastive loss** to align user descriptions with structured movie metadata.
- Outperformed a **GPT-4 few-shot baseline** on **Recall@K** and **MRR**, with strong retrieval quality.
- **Tech:** PyTorch, BERT, Contrastive Learning, Information Retrieval

SKILLS

Languages	Python, SQL, Java, C++
LLM Systems	LLM Serving, RAG Pipelines, Multi-Agent Systems and Tools, Prompt Engineering
LLM Training	Supervised Fine-Tuning, Transformers, PyTorch DDP
ML Infrastructure	CUDA, FlashAttention, Quantization, GPU Memory Optimization
Frameworks	PyTorch, CUDA, Triton, Transformers, vLLM, LangChain, LangGraph
Systems & Cloud	Docker, Kubernetes, Redis, Vector Databases, AWS, GCP, OCI

TECHNICAL INTERESTS

LLM Infrastructure • Agentic Systems • Retrieval-Augmented Generation • LLM Evaluation • Model Serving
• Distributed ML Systems • Reasoning Benchmarks

EDUCATION

M.S. in Computer Science, University of Colorado Boulder	Aug 2024 – May 2026
<i>Courses:</i> NLP, Deep Natural Language Understanding, Datacenter-Scale Computing	(GPA: 4.0/4.0)
B.E. in Computer Science, BMS College of Engineering	Aug 2017 – Aug 2021
<i>Relevant Courses:</i> Algorithms, Databases, Operating Systems, Machine Learning	

RESEARCH EXPERIENCE

Responsible LLM Evaluation for Mental Health Applications Peer-reviewed (IEEE-EMBS 2025)

- Built large-scale **LLM evaluation pipelines** measuring demographic bias, subgroup performance, and robustness for mental-health classification across GPT-4 and LLaMA models.
- Implemented **in-context prompting** and **fairness-aware losses** to mitigate bias in low-resource settings.
- Developed diagnostic metrics to quantify subgroup performance, consistency, and bias across model variants.

Commonsense Reasoning with Logical Entailment Trees

- Designed structured benchmarks and evaluation pipelines to assess multi-step commonsense reasoning in LLMs.
- Analyzed failure modes of chain-of-thought and n-shot prompting under compositional reasoning tasks.

PUBLICATIONS

- **Junias, Obed***, et al. *Assessing Algorithmic Bias in Language-Based Depression Detection: A Comparison of DNN and LLM Approaches*. IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2025. (Peer-reviewed, Presented). <https://ieeexplore.ieee.org/abstract/document/11269509>
- **Junias, Obed**, and Maria Leonor Pacheco. *LOGICAL-COMMONSENSEQA: A Benchmark for Logical Commonsense Reasoning*. arXiv preprint arXiv:2601.16504, 2026. <https://arxiv.org/abs/2601.16504>